

# CO-2-ASSESSMENT-TOOL-2

## Higher-Order Thinking Skills (HOTS) Assignment Question

COURSECODE:CSA-0924-JAVA

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1. Design a collection-based model for a social network containing users, followers, posts and hashtags. Justify where List, Set and Map should be used and identify potential scalability concerns.

Answer:

### **AIM:**

To design a collection-based social network model using Java Collections Framework to manage users, followers, posts, and hashtags efficiently.

### **Problem Statement:**

**We need to store**

- **Users**
- **Followers**
- **Posts**
- **Hashtags**

**The system should**

- **Avoid duplicate followers**
- **Maintain post order**

- Search users quickly
- Find posts using hashtags

**Which collections should we use ?**

| S.no | DATA      | COLLECTION             | WHY?                               |
|------|-----------|------------------------|------------------------------------|
| 1.   | Users     | Map                    | Search user quickly using username |
| 2.   | Followers | Set                    | No duplicate followers             |
| 3.   | Posts     | List                   | Maintain posting order             |
| 4.   | Hashtags  | Map<String,List<Post>> | Quickly find posts using hashtag   |

**COLLECTIN DESIGN:**

**Social Network**

**Map**

**username -----> User**

**User**

**Name**

**Followers (Set)**

**Posts (List)**

**Post**

**Content**

## Hashtags

### Hashtag Map

#java -----> List<Post>

#coding -----> List<Post>

#AI -----> List<Post>

### Pseudocode:

START

Create Map for Users

Create User

Create Set for Followers

Create List for Posts

Create Map for Hashtags

Add followers

Add posts

Store hashtags

Display user details

Display followers

Display posts

Search posts using hashtag

STOP

## CODE:

```
import java.util.*;

class Post {
    String content;

    Post(String content) {
        this.content = content;
    }

    public String toString() {
        return content;
    }
}

class User {
    String name;
    Set<String> followers = new HashSet<>();
    List<Post> posts = new ArrayList<>();

    User(String name) {
        this.name = name;
    }
}

public class SocialNetwork {
    public static void main(String[] args) {
```

```
// Users
Map<String, User> users = new HashMap<>();

// Hashtags
Map<String, List<Post>> hashtags = new HashMap<>();

User u = new User("Sakthi");
users.put("sakthi", u);

// Followers
u.followers.add("Rahul");
u.followers.add("John");
u.followers.add("Rahul"); // Duplicate ignored

// Posts
Post p1 = new Post("Learning Java #java");
Post p2 = new Post("AI Project #AI");
u.posts.add(p1);
u.posts.add(p2);

// Hashtags
hashtags.put("#java", new ArrayList<>());
hashtags.put("#AI", new ArrayList<>());
hashtags.get("#java").add(p1);
hashtags.get("#AI").add(p2);
```

```
// Display

System.out.println("User : " + u.name);

System.out.println("Followers : " + u.followers);

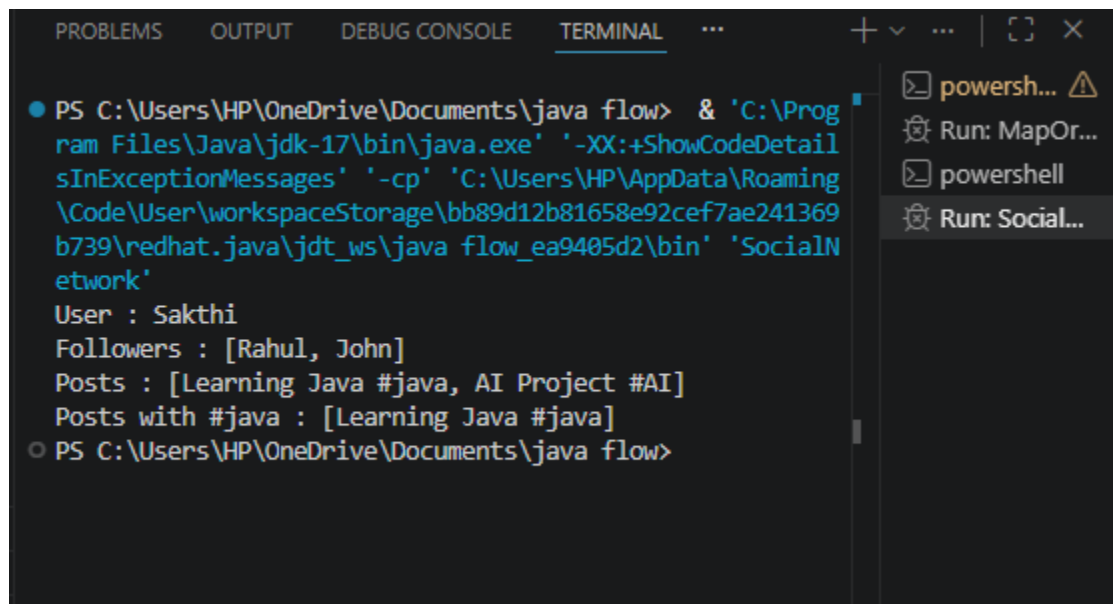
System.out.println("Posts : " + u.posts);

System.out.println("Posts with #java : " + hashtags.get("#java"));

}

}
```

## OUTPUT:



The screenshot shows an IDE terminal window with the following content:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  ...
● PS C:\Users\HP\OneDrive\Documents\java flow> & 'C:\Program Files\Java\jdk-17\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\HP\AppData\Roaming\Code\User\workspaceStorage\bb89d12b81658e92cef7ae241369b739\redhat.java\jdt_ws\java_flow_ea9405d2\bin' 'SocialNetwork'
User : Sakthi
Followers : [Rahul, John]
Posts : [Learning Java #java, AI Project #AI]
Posts with #java : [Learning Java #java]
○ PS C:\Users\HP\OneDrive\Documents\java flow>
```

On the right side of the terminal, there is a dropdown menu with the following options:

- powersh... ⚠
- Run: MapOr...
- powershell
- Run: Social...